INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 2: Literature Review and Theoretical Framework

by

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2 Literature Review and Theoretical Framework

The research problem set out in Chapter 1 is that a rollup holds far more contracts than anyone can inspect, and that the usual ways of forming trust in a contract do not read the allowances and transfers between contracts. The thesis asks whether a coupled PageRank over the allowance and transfer edges between contracts, C-PR, finds the contracts that users rely on, as a PageRank ordering was found to reach important web pages sooner than other orderings (Cho et al., 1998). Its reference walks are EndorseRank, over the allowance edges alone, and Adaptive Weighted PageRank (AWP), over the transfer edges alone; the objectives O1–O5 and research questions RQ1–RQ5 follow from that question. The gaps the literature leaves are taken up in Chapters 3 and 4.

2.1 Blockchain and DeFi foundations

A public blockchain records transactions in hashed blocks, and each block is secured by the ones added after it (Nakamoto, 2008). Once a majority of the nodes has accepted a block, rewriting it is costly, so parties can settle on the ledger without a known legal identity (*trust-minimized* settlement).

Bitcoin built an electronic cash network on such a ledger (Nakamoto, 2008); Ethereum lets an account hold a program alongside its balance. A transaction can then call one of the program's functions and change the contract's state (Buterin, 2014), and these functions run on the Ethereum Virtual Machine (EVM) (Wood, 2025). Such programs are called smart contracts (Szabo, 1997); any honest node that replays a function call gets the same output (Wood, 2025). Financial terms can therefore be enforced by the contract rather than by a trusted third party (Schär, 2021).

Together these contracts make up Decentralized Finance (DeFi), doing the work of exchanges,

lending platforms, derivatives markets and asset managers; Werner et al. (2022) define the protocol classes, and Schär (2021) describes how protocols are layered on one another. Because protocols interact, new financial instruments can be composed from existing ones (Kitzler et al., 2023), and contracts come to grant one another allowances and send one another transfers, the relations that the graphs of this thesis record. The same property lets a failure spread across protocols, as in flash-loan exploits (Qin et al., 2021) and price-oracle manipulation (Zhou et al., 2023). Transaction histories on these chains are public, yet the addresses in them are pseudonymous, which hampers the identity checks of conventional finance (D. Lin et al., 2025).

2.1.1 Pseudonymity and the choice of contracts

On a public blockchain the basic unit of analysis is an address (a wallet), which may stand for a legal entity, a trust, an algorithmic agent or a smart contract. One entity can own several addresses, and statistical clustering can recover ownership only in part (Victor, 2020). One consequence concerns credit: with no legal entity to tie a credit history to, on-chain lending asks for collateral worth more than the loan (Bartoletti et al., 2021), and proposed on-chain credit scores estimate repayment from on-chain behavior, with prospect theory (Hassija et al., 2020) or machine learning on lending data (Palaiokrassas et al., 2024).

A second, more basic consequence is the one this thesis takes up. Anyone can deploy a contract, an address says nothing about the code behind it, and the ledger does not say which of its very many contracts matter. A user asked to approve a spender, an explorer that labels addresses, an auditor and an integrator must each decide which contracts to look at first (Section 2.4).

2.2 Ethereum, Layer-2 rollups, and Arbitrum One

Most DeFi applications run on the Ethereum mainnet, which under proof of work could not raise its block size or shorten its block interval far without weakening its security (Gervais et al., 2016), so growing use raised gas costs and constrained throughput. Layer-2 (L2) protocols try to keep

Ethereum's security guarantees while making transactions cheaper: some execute transactions off-chain, others batch them and post a succinct summary to the mainnet (Gudgeon et al., 2020), and Thibault et al. (2022) survey and compare rollups. Optimistic L2s such as Arbitrum treat each submitted batch as valid unless it is challenged within a dispute period (Kalodner et al., 2018).

Arbitrum One is such an optimistic L2 and executes EVM smart contracts. We use it because the public BigQuery dataset includes both its ERC-20 `Approval` and `Transfer` events, and because its fees are low enough that the windows studied contain dense activity. It also holds very many contracts: Table 8.4 counts those in the five-token events of this study alone. We make no claim that our results generalize to other L2s or to Ethereum mainnet without further validation. The framework can be applied to any chain that emits the same logs.

2.3 ERC-20 tokens and the allowance lifecycle

The ERC-20 standard defines a common interface for fungible tokens, made up of `transfer`, `approve`, `transferFrom`, `allowance` and the associated events (Vogelsteller & Buterin, 2015). After the owner calls `approve(spender,amount)`, the token contract records that the spender may move up to `amount` out of the owner's balance. The spender can then move the tokens with `transferFrom`. With EIP-2612 `permit`, an allowance can be updated through an off-chain signature that settles in a single transaction, with the same semantics of authorization (Lundfall, 2020).

Figure 2.1 traces an ERC-20 allowance through its lifecycle. An allowance is an authorization and moves no tokens. An owner, whether an externally owned account or a contract, can grant allowances to routers, vaults or trading contracts that it trusts to act within set limits. Revoking (approving zero) and updating (setting a new non-zero allowance) both change the allowance currently in force, and `transferFrom` reduces it; some tokens record the remainder in a new `Approval` event. EndorseRank keeps only the most recent approval of each owner, spender and token, and that approval loses weight as it ages even while it stays in force (Section 2.8).

Between contracts, allowances are much sparser than transfers (Table 4.1): a single approval can

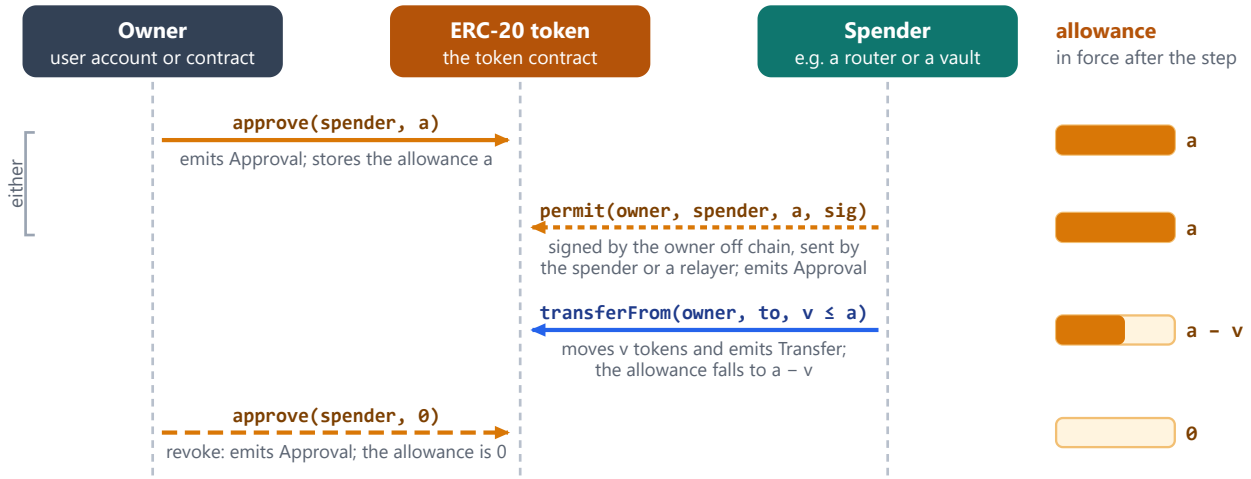


Figure 2.1: ERC-20 allowance lifecycle: `approve`, or its signature-based alternative `permit` (dotted), an optional `transferFrom`, and revoke. Orange arrows set the allowance and the blue arrow moves tokens; the bars on the right show the allowance in force after each step. EndorseRank keeps the most recent `Approval` event of each owner, spender and token in the score window and weights it by its age and by its US-dollar amount up to the owner's balance.

cover many transfers, most contracts approve no other contract on the five tokens, and most allowances that contracts do grant come from owners that hold none of the token at the freeze (Table 8.5).

2.4 Graph-based on-chain reputation

PageRank and its variants rank nodes by the influence that flows to them along directed edges (Page et al., 1999), and recent studies apply them to blockchain transaction graphs. H.-D. Do and Do (2023) run PageRank and HodgeRank on a sample of Ethereum transactions as a measure of social credit; their nodes are wallet addresses, whatever kind of account lies behind them, and the highest-ranked addresses they identify are exchanges, routers, token contracts and other protocol contracts. T. Do et al. (2023) propose Adaptive Weighted PageRank (AWP), which scores on-chain entities from their transaction histories and is evaluated here beside EndorseRank (Section 2.19). D. Lin et al. (2025) propagate risk through the transaction network to rate accounts without labels, and Wu et al. (2022) detect phishing addresses from a network embedding of the transaction graph.

Bellini et al. (2020) survey blockchain-based trust and reputation management, and Almasoud et al. (2020) review reputation systems built on smart contracts, in which users rate providers. Data mining on cryptocurrency transactions likewise works from transaction histories (Liu et al., 2021). In graphs of this kind every address in the data is a node. Mukhia et al. (2026) sort more than 81 million ERC-20 transfers on Ethereum by whether sender and recipient are externally owned accounts (EOAs) or smart contracts, and find that transfers between smart contracts follow other statistical patterns than those of EOAs. In the events of this study, user accounts (EOAs and smart accounts; Section 2.17) are most of the addresses and most of the accounts that newly approve a spender contract or send it a transfer (Tables 8.4 and 8.6). A user account costs no deployment, and the study does not measure who controls one, so the graphs of this thesis hold contracts only.

Outside the research literature, trust in a contract is formed in ways that each look at one contract or one account at a time. A wallet application shows a prompt before a user approves a spender; of the interfaces of 22 decentralized applications and 9 wallet applications that Wang et al. (2022) examined, 3 explained the approval mechanism. Code analysis reads the contract itself (Z. Lin et al., 2024); audit reports, which the incident study of Zhou et al. (2023) draws on, record a review of its code; and labels name known addresses but are sparse (D. Lin et al., 2024).

Attestation tooling summarizes evidence about the holder of an address. Human Passport (the former Gitcoin Passport) collects weighted verifiable credentials, called stamps, for checks such as government identity, biometrics, vouching and past web2 and web3 activity, and an address passes when the weighted sum reaches a threshold (Human Passport, n.d.), so it works as a gate and not as a graded ordering. None of these ways of forming trust, as reviewed here, uses the record of the allowances and transfers between contracts; the scores of this thesis are built from that record alone (Table 2.1).

A propagated score must also beat a simpler one. Where degree correlations are weak, as on the Web graph, PageRank is on average proportional to in-degree and can be approximated from it (Fortunato et al., 2008). New links concentrate on nodes that already have many, as preferential attachment assumes (Barabási & Albert, 1999) and measured network growth shows (Newman,

2001), so link prediction, on Ethereum transaction networks too (D. Lin et al., 2022), uses degree-based predictors as standard baselines (Liben-Nowell & Kleinberg, 2007).

2.5 PageRank formalism

Consider a directed graph $G = (V, E)$ with $|V|$ nodes, and write $w_{ij} > 0$ for the weight of the edge from i to j. That weight stands for influence from i to j, in the endorsement or the transfer sense. Let W be the corresponding weighted adjacency matrix, and let $o_i = \sum_j w_{ij}$ be the out-strength of node i. The row-stochastic transition matrix P is then

$$P_{ij} = \begin{cases} w_{ij}/o_i & \text{if } o_i > 0, \\ 1/|V| & \text{if } o_i = 0 \end{cases} \tag{2.1}$$

(dangling nodes spread their mass evenly). Given a damping parameter $d \in (0, 1)$, the Google matrix is

$$G_{\text{PR}} = d\, P^{\top} + \frac{1-d}{|V|}\, \mathbf{1}\mathbf{1}^{\top}, \tag{2.2}$$

and the PageRank vector $\mathbf{r}$ satisfies $\mathbf{r} = G_{\text{PR}}\mathbf{r}$ with $\mathbf{r} \geq 0$ and $\sum_i r_i = 1$ (Langville & Meyer, 2006). Element by element, with the dangling rows of P written as a separate middle term, this becomes

$$r_j = d \sum_{i:(i\to j)\in E} r_i \frac{w_{ij}}{o_i} + \frac{d}{|V|} \sum_{i:\, o_i = 0} r_i + \frac{1-d}{|V|}. \tag{2.3}$$

2.5.1 Random-surfer interpretation

Equation (2.3) can be read probabilistically. At each step a surfer at a node with out-edges follows one of them with probability d, choosing it in proportion to its weight, or teleports to a uniformly chosen node with probability $1 - d$ (Figure 2.2). At a node without out-edges the surfer always jumps to a uniformly chosen node; this is the dangling rule of Equation (2.1). The stationary distribution then ranks nodes by how often they are visited in the long run. Damping keeps rank

sinks (groups of nodes with no edge leaving them) from absorbing all the mass. With $d < 1$ and uniform teleportation, every entry of G_{PR} is positive, so G_{PR} is primitive and its Perron vector is unique (Langville & Meyer, 2006). Appendix 8.3.1 gives the argument.

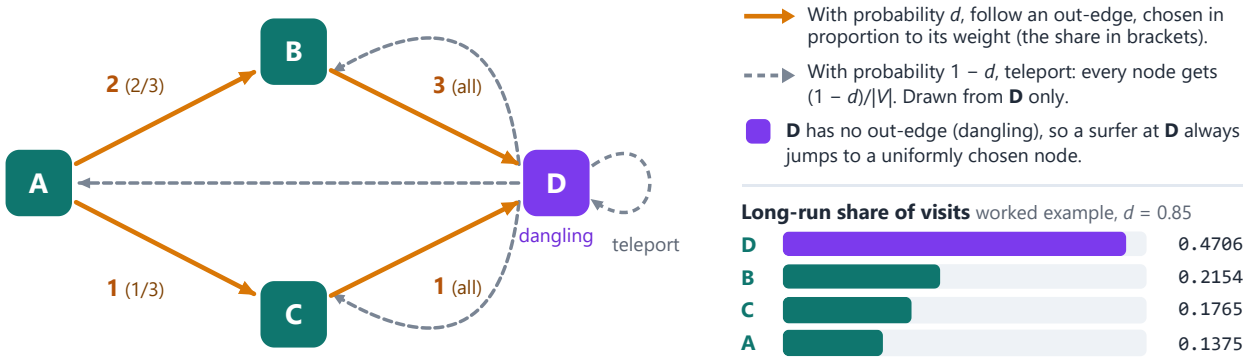


Figure 2.2: Random-surfer intuition for PageRank on the four-node graph of Section 2.14; edge labels are the allowance weights, with the share of a step that each edge takes in brackets. Solid arrows are graph transitions, followed with total probability d in proportion to their weights. Dashed arrows show uniform teleportation, which reaches every node with probability $(1-d)/|V|$; the dashed fan is drawn only from D, although every node has one. D has no outgoing edge, so under the dangling rule of Equation (2.1) a surfer at D always jumps to a uniformly chosen node. The bars give the stationary vector of the worked example, the long-run share of visits.

2.5.2 Power iteration and complexity

In practice, $\mathbf{r}$ is usually obtained by power iteration from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/|V|$,

$$\mathbf{r}^{(t+1)} = G_{\mathrm{PR}}\mathbf{r}^{(t)}, \tag{2.4}$$

until $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or until the iteration cap $I_{\max}$ is reached. On a sparse graph, each iteration costs $O(|E|)$ arithmetic operations (Leskovec et al., 2020). A method with fewer edges, such as latest-allowance snapshots in place of full transfer histories, therefore lowers the cost per iteration and sometimes the memory.

By default this work sets damping to the conventional $d = 0.85$ (Page et al., 1999) and uses two study-specific solver settings, a convergence tolerance $\varepsilon = 10^{-8}$ and a maximum of $I_{\max} = 300$

iterations.[1] Damping is varied over $d \in \{0.75, 0.85, 0.95\}$ in the robustness section.

2.6 Crawl ordering

A crawler has read only some pages and must choose which of the pages they link to it reads next; if it stops early, that order decides which pages it holds. Cho et al. (1998) study this problem of URL ordering: to visit the important pages first with an ordering that uses only what the crawl has seen. Importance is, among other measures, a page's backlink count over the whole collection or its PageRank; the orderings are breadth-first order and the same two quantities computed on the pages seen so far; and an ordering is judged by the share of the *hot* pages, those above an importance threshold, visited after a given number of pages. On a stored copy of the Stanford University web, about 179,000 pages reachable from the home page where every crawl began, the crawl ordered by PageRank (damping factor 0.9) reached the pages with at least 100 backlinks sooner than the crawl ordered by the backlink count. The authors call the result counter-intuitive and trace it to the start of a crawl, where the backlink count keeps the crawler inside one cluster of pages. With hot pages defined by PageRank, the PageRank ordering led by more. The authors name the single collection as a limitation.

Najork and Wiener (2001) crawled 328 million pages in breadth-first order, found that pages with a high PageRank were downloaded early, and argue that this order is the better trade-off because recomputing PageRank during such a crawl is too costly; they ran no other ordering. Baeza-Yates et al. (2005) simulated crawls of the Chilean and the Greek web, between two and four million pages each, with PageRank as the importance: orderings by the backlink count and by a partially updated PageRank did worse than a random crawl, breadth-first order was close to the best orderings only over the first 20 to 30 percent of the pages, and the authors place PageRank recomputed in batches on the pages seen so far among the best, together with two cheaper strategies that they recommend. In both studies that compare orderings, the backlink count thus trails a PageRank that is recomputed

[1] They keep run-times reproducible and are not part of the classical PageRank formulation.

in full on the pages seen so far.

The discovery-order test of this thesis repeats the experiment of Cho et al. (1998) on the contract graph (Section 3.7). A page becomes a contract and a link an allowance or transfer edge between two contracts; a visit reads a contract's edges, and the next contract is chosen by an ordering computed on what has been read so far: breadth-first order, the number of contract backlinks, or a walk such as C-PR. Importance is taken from outside the graph: it is the demand of user accounts, the US dollars they transferred to a contract and the balance-capped US dollars they authorized it to spend, which no ordering reads. Importance by backlink count and by PageRank, as in the original experiment, is an intended-construct check and is reported second. Figure 2.3 shows the experiment and its counterpart on the contract graph.

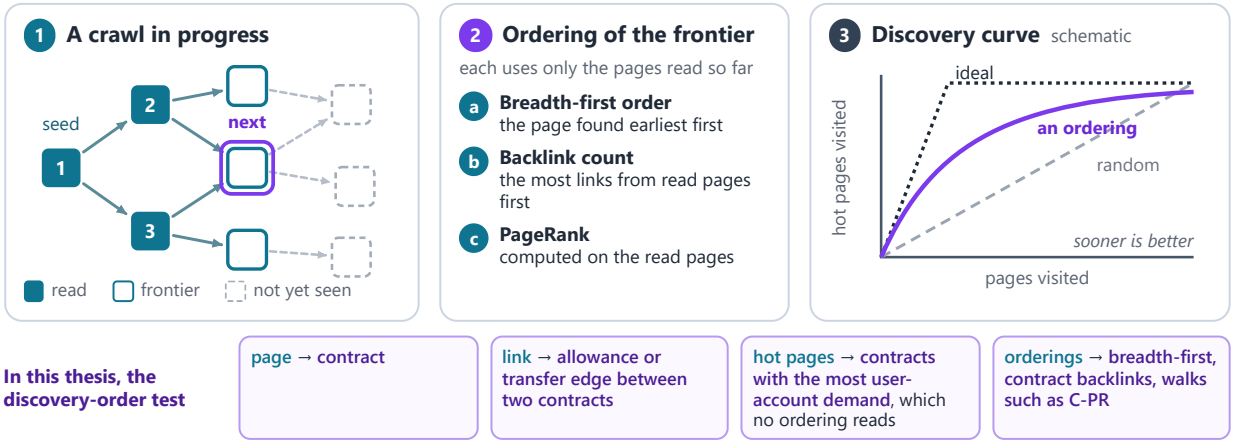


Figure 2.3: The crawl-ordering experiment of Cho et al. (1998), drawn as a schematic without data. A crawl that began at a seed page has read some pages (filled, numbered in the order of reading); the frontier holds the pages they link to, and an ordering computed on the pages read so far chooses the next one. An ordering is judged by how soon it visits the hot pages, those above an importance threshold, between a random order and the ideal one. The bottom row gives the counterparts in the discovery-order test of this thesis (Section 3.7).

2.7 Reputation systems lineage

PageRank belongs to a wider family of spectral and propagation-based reputation mechanisms. Kleinberg's HITS algorithm gives each page a hub score and an authority score (Kleinberg, 1999). EigenTrust extends the idea to peer-to-peer networks and normalizes local trust, so a malicious node

cannot raise its own score at will (Kamvar et al., 2003). To suppress web spam, TrustRank starts from a seed set of trusted nodes and propagates their trust outward (Gyöngyi et al., 2004). Where trust flows only from a vetted seed set, a newcomer cannot raise its score without endorsements from nodes that already hold reputation. With uniform teleportation, which EndorseRank uses (Section 3.6.2), that protection is weaker.

Figure 2.4 places the ranking methods reviewed in this chapter by family and year, together with what the walks of this thesis take from each family.

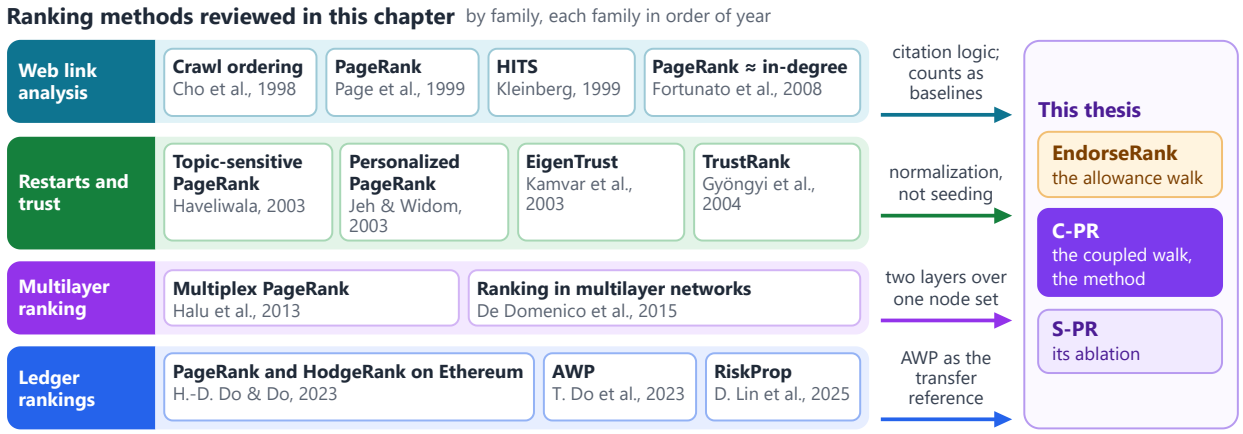


Figure 2.4: Ranking methods reviewed in this chapter, by family and, within a family, by year, with what the walks of this thesis take from each family (arrows). The ledger rankings take every address in their data as a node; the walks of this thesis take contracts only (Section 2.4).

2.8 ERC-20 allowances as endorsement

Through the ERC-20 `approve` function (Vogelsteller & Buterin, 2015) and EIP-2612 `permit` (Lundfall, 2020), an owner states explicitly how much a spender may move on its behalf. EndorseRank reads the latest allowance from an owner contract to a spender contract as one endorsement. On each of the five tokens of the study (Section 3.5.3), the latest in-window approval of the pair contributes its time weight times the bounded value weight of its US-dollar amount, as AWP weights a transfer; a pair whose latest value is zero is no edge, and the terms are summed over tokens (Equation (3.7) in Section 3.6.2). PageRank on the resulting graph G_{approve} yields the

EndorseRank scores $\mathbf{s}_{ER}$.

An allowance may exceed the owner's balance, and an unlimited one always does, yet a spender can never move more than the owner holds. Each latest allowance therefore counts up to its owner's balance of the token at the freeze, in US dollars at the price of that day, and an allowance whose owner holds none of the token is no edge (the balance cap; Section 3.6).

Treating allowances as endorsements is a deliberate choice, and an imprecise one. An allowance between two contracts is a working relation between programs, not a judgment that the spender is trustworthy, and an allowance manager such as Uniswap's Permit2, which owners approve once so that applications can spend through it, collects the endorsements meant for those applications (Section 3.6.2). Even so, an allowance signals approval more explicitly than a received transfer, which could come from market making, wash trading (Cong et al., 2023), airdrop farming (Luo et al., 2025) or routing without any intent to endorse.

Victor's self-authorization heuristic reads ERC-20 approvals as a relation between addresses and clusters those that it attributes to one entity (Victor, 2020). That reading links addresses and ranks nothing. It is also a caution, since an approval between two addresses of one entity is a self-endorsement.

2.9 What this research does not claim

The evidence in this thesis is of three kinds, and none of them an outcome for the users of a contract. Importance in the discovery-order test is the demand of user accounts; it shows that users rely on a contract, not that they are right to. The out-of-window labels count later events of the two kinds from which the graphs are built. Construct validity expects a score to agree with checks of the construct it was built to measure (Cronbach & Meehl, 1955), so each walk's lead on the same-window checks of its own edges is partly built in. No score in this thesis is offered as a measure of the safety of a contract, of the gains or losses of its users, or of whether a party will repay what it owes. Nor is any score a security audit: a contract can contain malicious functions whatever

its standing among owners (Z. Lin et al., 2024), and the scores read neither code nor upgrade permissions. Apart from the token list and daily prices, the scores use no off-chain information.

2.10 Rank correlation theory

Reputation scores and the proxies of the construct checks are ordered quantities without a natural boundary for classification. Following the AWP validation design (T. Do et al., 2023) and classical psychometrics (Cronbach & Meehl, 1955), this research measures two kinds of rank agreement, monotonic and pairwise. Bootstrap confidence intervals quantify the sampling uncertainty of every coefficient of the main analysis (Efron, 1979).

Spearman's rank correlation (Spearman, 1904) is the Pearson correlation of the ranks. For paired ranks (R_i, S_i) of n items without ties, it reduces to

$$\rho = 1 - \frac{6 \sum_{i=1}^{n} (R_i - S_i)^2}{n(n^2 - 1)}. \tag{2.5}$$

With ties, Spearman's ρ is the Pearson correlation of average ranks (mid-ranks), in which tied items share the mean of their positions, and every reported ρ is computed this way.

Kendall's τ (Kendall, 1938) rests on counts of concordant and discordant pairs. Of the $n_0 = n(n-1)/2$ pairs (i, j) with $i < j$, n_c are concordant (both rankings order them the same way) and n_d are discordant (the rankings order them oppositely); a pair tied in either ranking is neither. Let $n_1 = \sum_k t_k(t_k - 1)/2$, where t_k is the size of the k-th group of tied values in the first ranking, and define n_2 in the same way for the second ranking. The τ-b variant is then

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_0 - n_1)(n_0 - n_2)}}, \tag{2.6}$$

and without the tie corrections it becomes $\tau_a = (n_c - n_d)/n_0$. Every τ reported in this thesis is τ_b, computed on the raw scores and proxies. The difference matters for heavily tied scores: the raw degrees are integer counts, and a walk gives one value to every contract that has no edge in its

graph, which for EndorseRank is most of the matched cohort (Section 4.4). Kendall's τ is usually the easier coefficient to interpret for pairwise orderings, and Chapter 4 uses it as the main family-level summary. Its values are compared with those of other methods on the same cohort; no uncited absolute benchmark is used.

Figure 2.5 works through these counts on four contracts with made-up values.

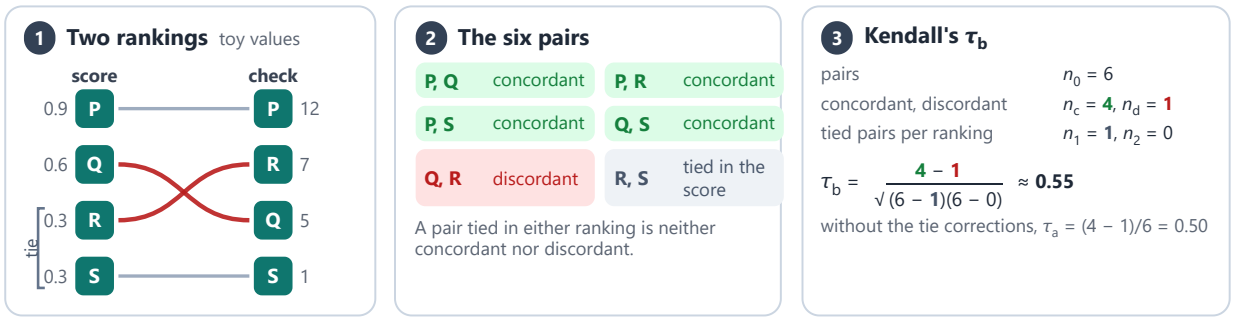


Figure 2.5: Concordant and discordant pairs and Kendall's τ_b (Equation (2.6)) on four contracts with made-up values. The lines of Q and R cross, so that pair is discordant; R and S tie in the score, so their pair is neither concordant nor discordant and enters n_1.

2.11 Theoretical framework

The study first compares the two single-layer walks, EndorseRank and AWP, on the same contracts: whether they order them differently (H1), whether each agrees with the checks of its own edge type (H2) and what each costs (H3); together these make up RQ1. It then tests the coupled walk C-PR, the method of the thesis, by the order in which a crawl that follows it reaches the contracts with the most user-account demand (RQ2), by the prediction of later approvals and senders against the raw counts (RQ3), and under its rules (RQ4 and RQ5). RQ4 and RQ5 follow the rules of the analysis plan and of the registration, applied to the scores of the contract graph as a re-analysis; RQ2 and RQ3 follow criteria defined after the registered results were known (Section 3.5.3). Six arguments frame these steps.

- Hyperlink–citation semantics: In the original PageRank algorithm (Page et al., 1999), an inbound hyperlink counts as a citation. H.-D. Do and Do (2023) apply this reasoning to

Ethereum transactions, and AWP (T. Do et al., 2023) weights each transfer edge by the age and the value of its transfers (Section 3.6.1). EndorseRank extends citation-based propagation to ERC-20 approve and permit edges: an allowance is an explicit, on-chain delegation of spending authority (Vogelsteller & Buterin, 2015).

- Contracts as nodes: User accounts cost nothing to create, and they supply most of the new counterparties that contracts gain (Table 8.6), so their edges would dominate a walk over all addresses. A contract costs a deployment, and its allowance counts only up to the balance it holds, so the graphs hold contracts only and no score or baseline reads an edge of a user account. User accounts enter only through the labels and as the importance that the discovery-order test asks an ordering to find (Section 2.6).
- Domain alignment framework: Measurement theory uses construct alignment to ask how closely an indicator follows the construct it is meant to track (Cronbach & Meehl, 1955), and Messick (1995) widens the idea into one unified judgment about what a score means and what using it implies. Graph coherence is checked with rank correlations (Section 2.10) against same-window transfer and allowance statistics; the out-of-window checks are temporal holdouts on the spender cohorts, new approval pairs and new transfer senders in W0 and in the registered window W1.
- Computational complexity: Each PageRank iteration costs $O(|E|)$ (Section 2.5.2), so on the sparser allowance graph an iteration needs less time and memory; this argument underlies H3. The number of iterations also depends on the damping factor (Langville & Meyer, 2006), the graph and the restart vector. Every walk uses the same solver, so runtime differences come from the graphs and restart vectors and are not an algorithmic contribution (Appendix 8.3.2).
- Local counts as a baseline: Where contracts that already have many counterparties gain the most new ones (Section 2.4), the raw count is a strong predictor of future arrivals, and in the crawl-ordering experiments PageRank is measured against the backlink count. A PageRank differs from the count in three ways: the endorsers' own scores propagate along the edges, the

edges are weighted by age and amount, and the walk restarts by a rule of its own. RQ2 and RQ3 therefore set each walk against the count of the edges it smooths, and neither attributes a difference to propagation alone.

- Layer coupling: Multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) treat several relations as layers over one node set. Mixing the layers node by node, as C-PR does, leaves a question open for layers of very different density: a node with no out-edge in the sparse layer follows the dense one, so the signal of the sparse layer may or may not survive. RQ4 therefore asks whether the mixture beats both layers (rule A), and RQ5 whether it improves on its transfer layer walked alone (two of rule B's registered criteria, F1 and F2; Section 3.8.1). Both layers weight edges by US-dollar amounts without AWP's bounded value weight and are walked with uniform restarts, so neither walk is AWP or EndorseRank, and C-PR is compared with AWP separately (Section 4.8.4). Coverage matters there: a walk over one layer leaves every contract without an edge of that layer in one tie, and the other layer may order some of them.

Figure 2.6 sets out these parts. Only EndorseRank and AWP enter H1; H2 and H3 concern every score.

2.12 Terminology

The formal constructs are defined in Chapter 1; the terms below are used in interpreting results:

- Contract and user account: every node of every graph is a contract; a user account (an EOA, an EIP-7702 account or a smart account) is a node of no graph (Section 2.17).
- Single-layer walks: EndorseRank (allowance layer) and AWP (transfer layer), the two PageRank scores that walk on one edge type; C-PR walks both (Section 3.6.3).
- Balance cap: an allowance counts up to its owner's balance of the token at the freeze.
- Intended-construct check: agreement of a score with a same-window statistic of its own edges (Cronbach & Meehl, 1955); it is not offered as validation.

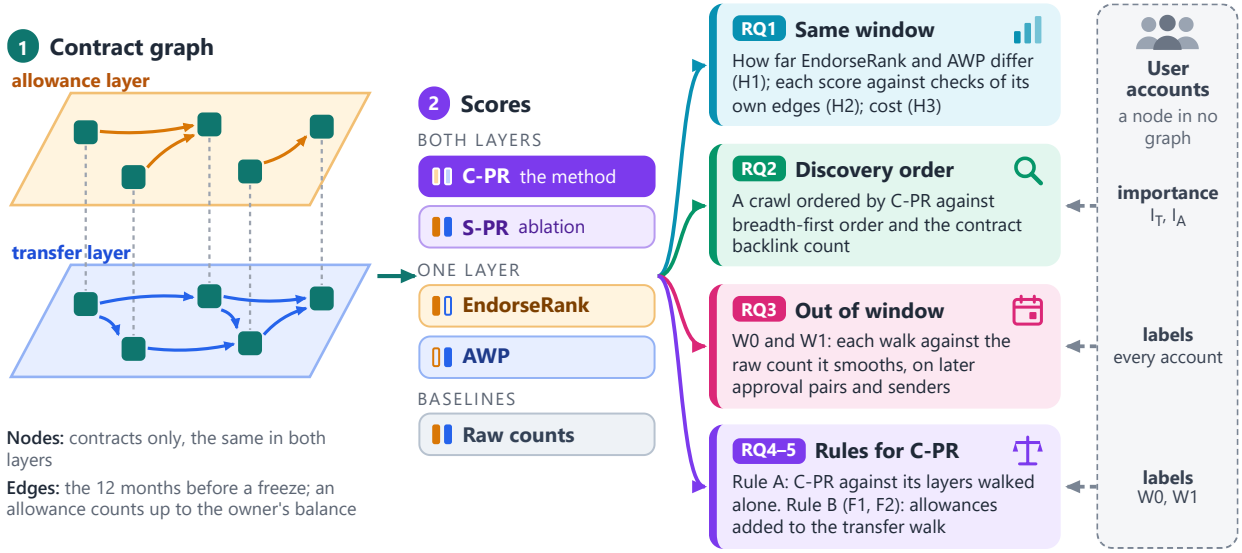


Figure 2.6: Conceptual model. The contract graph of a freeze has two layers, allowances and transfers, over the same contract nodes, and it is the only input of the scores; the bars beside a score mark the layers it reads. Every score enters the four evaluations, from top to bottom in the order of the research questions. User accounts (dashed) are nodes of no graph: they enter only as the importance of the discovery-order test and through the out-of-window labels, which count every account.

- Discovery-order test: crawls of the contract graph, each ordered by a score computed on the part read so far, compared by how soon they reach the contracts with the most user-account demand (Section 3.7).
- Out-of-window prediction (temporal holdout): in the first window, W0, scores are frozen at t_1 (31 March 2026), and new owner–spender approval pairs and new transfer senders are counted up to t_2 (30 June 2026); a registered second window, W1, freezes scores on 30 June 2026 and counts labels from July to September 2026 (Section 3.8.1). The matched cohort and the spender cohorts are the contract sets of Section 4.1.1.

2.13 Domain alignment metrics

Construct alignment is measured by Spearman's ρ and Kendall's τ between scores and check rankings (Section 2.10), on same-window checks and on the out-of-window labels; the discovery-order test has a statistic of its own (Section 3.7). Chapter 3 gives the operational definitions.

2.14 Worked example: PageRank on a four-node endorsement graph

Take four contracts $\{A, B, C, D\}$ joined by the latest-allowance edges

$$A \to B\ (2), \quad A \to C\ (1), \quad B \to D\ (3), \quad C \to D\ (1),$$

with damping $d = 0.85$. The out-strengths are $o_A = 3$, $o_B = 3$, $o_C = 1$ and $o_D = 0$ (the last is dangling). The edges $B \to D$ and $C \to D$ are the only out-edges of their nodes, so each is followed with probability one and its weight has no effect on the scores. Teleportation adds $(1 - d)/4 = 0.0375$ to every node at every step, the first one included. One power-iteration step from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/4$ gives $\mathbf{r}^{(1)} = (0.0906,\ 0.2323,\ 0.1615,\ 0.5156)$ for (A, B, C, D). The iteration converges to the stationary vector $\mathbf{r} = (0.1375,\ 0.2154,\ 0.1765,\ 0.4706)$, so D comes first, B and C follow, and A is last. No contract endorses A. Apart from teleportation, its whole score comes from the dangling rule, which spreads D's mass evenly over the four nodes ($d\, r_D/4 = 0.1000$). The example shows that a node gains score by *receiving* endorsements from nodes that are endorsed in turn, which is the citation logic of PageRank (Page et al., 1999).

Transfers among the same contracts would in general join other pairs, or the same pairs with other totals, and so give another transition matrix P and possibly a different ranking. Hypothesis H1 is this situation in miniature: EndorseRank and AWP need not agree. They also restart differently, so H1 compares edges and restarts together; Section 4.5.4 walks the allowance graph with AWP's restarts.

2.15 DeFi risk taxonomy and where reputation fits

Werner et al. (2022) separate the technical security of DeFi protocols from their economic security, and Zhou et al. (2023) place the attacks and accidents seen in practice in one reference frame. Bugs in contract code are smart-contract risks (Li et al., 2022); a stale price, or a feed under someone's control, is an oracle risk (Angeris & Chitra, 2020); volatility and liquidation cascades are market

risks (Perez et al., 2021); and voting power held by a few addresses is a governance risk (Fritsch et al., 2024). Each layer starts from a given protocol and asks what can go wrong with it. The thesis asks the question that comes first: among very many contracts, which ones users rely on. A ranking of contracts by the relations between them is no substitute for auditing or oracle design; it is meant to say where such methods should be applied first, not to predict exploits or governance attacks. Figure 2.7 sets this question before the risk layers.

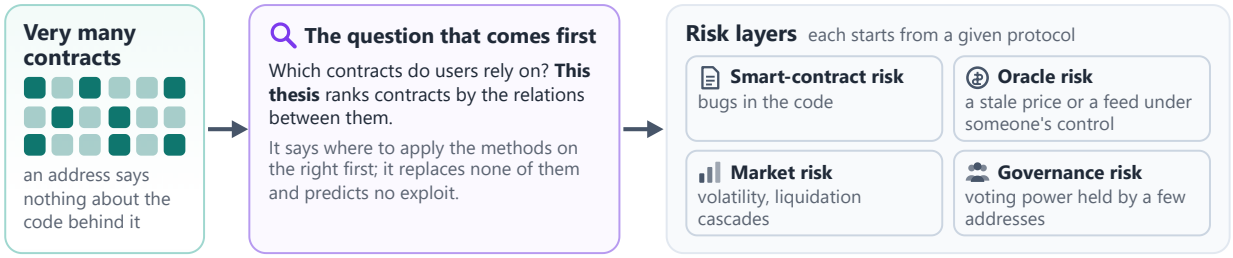


Figure 2.7: Where a ranking of contracts fits beside the DeFi risk layers. Each risk layer starts from a given protocol and asks what can go wrong with it; the ranking asks first which of very many contracts users rely on, to say where those methods should be applied first. It replaces none of them and predicts no exploit.

Artificial activity makes any ranking from ledger data harder to trust. Wash trading inflates volume on decentralized exchanges (Victor & Weintraud, 2021) and, on a larger scale, on unregulated centralized exchanges (Cong et al., 2023), extractable value creates transfers that follow ordering strategies rather than relationships (Daian et al., 2020), and scam tokens add transactions without sustained relationships (Xia et al., 2021). The transfer layer is sensitive to such volume, and the study does not measure it. The allowance layer is sensitive to approvals granted only to raise a score; the contract-only graph and the balance cap raise their cost, and Section 4.1 checks the graphs for closed groups of contracts, which would keep the mass of a walk.

2.16 Related measurement traditions

Outside blockchains, reputation systems have been studied in e-commerce (Resnick & Zeckhauser, 2002), peer-to-peer networks (Kamvar et al., 2003) and web spam (Gyöngyi et al., 2004). Three insights from this work apply here: (i) reputation is a property of relations; (ii) seeding and nor-

malization limit the effect of inflation; and (iii) evaluation needs external benchmarks, because internal coherence alone is not enough (Cronbach & Meehl, 1955). The thesis takes up (i) directly. Of (ii) it takes up normalization but not seeding (Section 2.7). On (iii) it goes part of the way: the same-window checks test coherence, the out-of-window prediction tests later events of the same two kinds, and the discovery-order test uses a benchmark that no score reads.

Topic-sensitive PageRank (Haveliwala, 2003) and personalized PageRank (Jeh & Widom, 2003) bias the teleportation vector toward a chosen group of nodes, as TrustRank does with its seed set (Gyöngyi et al., 2004) and AWP with sending activity (T. Do et al., 2023). Multiplex PageRank lets a node's centrality on one layer shape the walk on another (Halu et al., 2013), and the multilayer framework of De Domenico et al. (2015) lets one walker move within and between the layers of an interconnected network. C-PR takes the simplest form of this idea, a per-node weighted sum of two transition matrices with one weight, and S-PR uses the allowance score as the teleportation vector (Section 3.6.3).

2.17 Accounts, gas, and why logs are the measurement surface

Ethereum has two kinds of account: externally owned accounts (EOAs) and contract accounts. A private key controls an EOA, whereas a contract account is governed by the code it stores (Antonopoulos & Wood, 2018). The line between a user's account and a contract does not follow this division exactly: an EOA can delegate to code under EIP-7702 (Buterin et al., 2024), and a smart account, such as a Safe (Safe Ecosystem Foundation, n.d.) or an ERC-4337 account (Buterin et al., 2021), is a contract account that acts for a user. This thesis therefore separates *user accounts* (EOAs, EIP-7702 accounts and smart accounts) from *contracts* (every other contract account; Section 3.5.1) and scores contracts only (Section 3.2). Every state-changing transaction consumes gas. Logs come from a contract's `LOG` instructions, which run when its code emits an event (the Solidity `emit` statement); writing to storage produces no log. A reputation system that means to be auditable must prefer logs to off-chain databases, because logs are a permanent part of

the blockchain history that any full node or index provider can recompute (Wood, 2025).

Figure 2.8 shows the two kinds of account and the logs that the study reads.

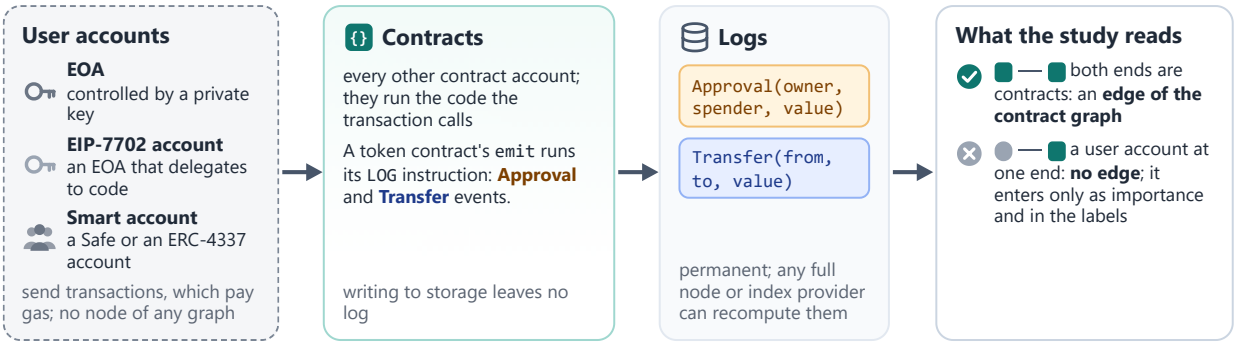


Figure 2.8: Accounts, transactions and the logs that the study reads. User accounts (dashed) send transactions, which pay gas; contracts run the code a transaction calls, and a token contract records `Approval` and `Transfer` events as logs, while writing to storage leaves none. An event whose two ends are contracts becomes an edge of the contract graph; an event with a user account at one end is no edge and enters only as importance and in the labels.

2.18 Approvals as security-relevant events

Unlimited approvals (`type(uint256).max`) save the owner from approving again, at a cost in safety. If the spender is compromised, the owner's whole balance of the token can be stolen until the approval is reset. An allowance is therefore a capability and not a record: whoever holds it can move the owner's tokens until it is spent or revoked (Vogelsteller & Buterin, 2015). Transaction-based phishing exploits this: fake websites lead users to sign transactions that let scammers take their assets (He et al., 2023).

Wang et al. (2022) find that unlimited approvals make up the majority of ERC-20 approvals on Ethereum and quantify the risk they carry. They are the majority of the positive latest allowances between the contracts of this study as well (Table 8.5). Under the balance cap such an approval counts for what its owner holds of the token at the freeze, like any allowance above the balance; in the allowance layer of C-PR a second cap, in US dollars, bounds what one large balance can add (Section 3.6.3).

2.19 AWP in depth: what the transfer-graph baseline measures

AWP, proposed by T. Do et al. (2023), ranks the addresses of a transaction graph. Each transaction adds to its edge a logistic function of its time multiplied by a bounded function of its value, so that no transaction adds more than one to the edge weight. The walk restarts at each address in proportion to its activeness, the sum over the addresses it has sent to of the time weight of its latest transaction to each. Ranking accuracy is then validated against inbound degree and inbound value, taken as proxies for the economic attention a wallet attracts. The form of the decay (Equation (3.3)) and the in-degree/in-value proxies in Chapters 3 and 4 are taken from the AWP paper. The paper gives no parameter values; Section 3.6.1 gives the values used here.

EndorseRank keeps AWP's weight functions under PageRank and its rank-correlation check, and changes what the edges mean and where the walk restarts: EndorseRank restarts uniformly, AWP at active senders (Section 3.6.2), which on the contract graph are contracts.

2.20 Comparison of graph reputation methods

Table 2.1 compares the scores of this thesis with the other approaches of this chapter by edge semantics, temporal model and validation design.

As Table 2.1 indicates, the contribution of the thesis lies in the graph, the operator and the evaluation. The graph holds contracts only and reads two relations between them, one of which, the allowance, the reviewed methods do not use as an edge. The operator, C-PR, couples the two relations into one walk, with a seeded ablation to measure how much the coupling contributes. The evaluation sets the walk against the demand of user accounts, against the raw counts and against its rules (Section 2.11). EndorseRank's contribution is the use of the latest allowance as the edge of a ranking; it is neither the ranking algorithm nor the reading of approvals as a link between addresses (Section 2.8).

Table 2.1: Comparison of selected graph reputation methods (conceptual).

Method	Edge semantics	Temporal model	Typical validation
PageRank (web) (Page et al., 1999)	Hyperlinks	None / static snapshot	Retrieval quality; crawl order (Cho et al., 1998)
EigenTrust (Kamvar et al., 2003)	Local trust ratings	Iterative aggregation	P2P file authenticity
TrustRank (Gyöngyi et al., 2004)	Links from seed goods	Static / seeded	Spam demotion
PageRank/ HodgeRank on Ethereum (H.-D. Do & Do, 2023)	Transactions among wallet addresses	Static sample	Rank correlation with transaction counts
AWP (T. Do et al., 2023)	Transactions, bounded value weights	Logistic decay of edges and restarts	In-degree / in-value
RiskProp (D. Lin et al., 2025)	Transactions, with a de-anonymization score	Propagation of risk	Labeled illicit accounts
Multiplex PageRank (Halu et al., 2013); multilayer ranking (De Domenico et al., 2015)	Several relations over one node set	Static layers	Versatility of top-ranked nodes
Human Passport (n.d.)	No edges; weighted credentials about the address holder, on-chain and off-chain	Current credential set	Admission threshold
EndorseRank (this study)	Latest allowances between contracts, up to the owner's balance; AWP's weight functions	Logistic decay of each latest approval; uniform restarts	Allowance checks; later approval pairs
C-PR / S-PR (this study)	Allowance edges mixed with, or seeding, transfer edges between contracts; allowances capped	Allowance snapshot and logistic decay of transfers	Discovery order; out-of-window labels; rule A and two criteria of rule B

2.21 Summary of the reviewed literature

Table 2.2 collects the sources underlying this study, with the strengths this thesis draws on and the limitations it tries to address.

Table 2.2: Summary of the reviewed literature: methodology, strengths, limitations, and relation to this study.

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Page et al. (1999); Langville and Meyer (2006)	Eigenvector ranking of a hyperlink graph, solved by power iteration	Interpretable; $O(\|E\|)$ per iteration; a unique Perron vector under damping	Edges mean hyperlinks; no notion of time	Same solver and complexity argument for every walk
Cho et al. (1998); Najork and Wiener (2001); Baeza-Yates et al. (2005)	Web crawls ordered by breadth-first order, backlink count or PageRank on the pages seen	Test an ordering under partial knowledge	Importance taken from the links themselves; results vary with the collection	Model of the discovery-order test, with user-account demand as importance
Kamvar et al. (2003); Gyöngyi et al. (2004)	Trust aggregation in P2P networks; propagation from seeds to demote spam	Normalization and seed sets curb inflation; explicit adversary	Need ratings or curated seeds; not applied on-chain	Inform the choice of uniform restarts
H.-D. Do and Do (2023)	PageRank and HodgeRank on Ethereum transactions, as social credit	Public data; the top-ranked addresses are exchanges and protocol contracts	Addresses of every kind in one graph; checked against transaction counts	Ledger PageRank over all addresses; here the graph holds contracts only
T. Do et al. (2023), AWP	Transaction PageRank with value weights, logistic recency and activity restarts	Validates on ranks; models recency	Proxies come from the same transfers as the graph; no parameter values reported	Transfer-layer reference; parameters set here (Section 3.6.1)

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
D. Lin et al. (2025); Wu et al. (2022)	Risk propagation and network embedding on Ethereum transaction graphs	Rate or detect accounts that carry no label	Need labels of illicit accounts; aimed at fraud	Show that graph position is informative; not reproduced here
Kitzler et al. (2023); Mukhia et al. (2026)	DeFi protocol networks; ERC-20 transfers by EOA and contract ends	Protocols interact mainly in one strongly connected component; contract-to-contract transfers differ	No ranking of contracts	Support a graph of contracts only
Hassija et al. (2020); Kotb (2023)	Credit scoring from blockchain data	Aimed at lending decisions	Need labels or off-chain features	Background; no such score is built here
Bellini et al. (2020); Almasoud et al. (2020)	Surveys of blockchain-based trust and reputation systems	Taxonomies of the design space	Reputation from users' ratings; no allowance edges; nothing measured on L2	None of the surveyed designs uses allowance edges
Victor (2020); Victor and Weintraud (2021); Cong et al. (2023)	Address clustering, partly through ERC-20 approvals; wash-trading detection	Measure multi-address and artificial activity	No ranking proposed	Closest prior use of approvals; transfer signals can be inflated
Human Passport (n.d.)	Weighted credentials about an address holder	In deployed use; adds off-chain evidence	A gate, not an ordering; relations between addresses play no part	How trust is formed in practice
Cronbach and Meehl (1955); Messick (1995)	Psychometric construct validity	Separate what a score is designed to measure from external criteria	Not specific to blockchains	Basis for reading same-window agreement as an intended-construct check

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Efron (1979); DiCiccio and Efron (1996)	Bootstrap resampling and confidence intervals	Uncertainty for rank statistics without distributional assumptions	Units must be exchangeable	Source of the 95% intervals and paired contrasts of Chapter 4

2.22 Research gap

Five gaps emerge from Table 2.2 and Sections 2.6 and 2.16. First, the ledger rankings reviewed here rank addresses in general from their transactions (PageRank and HodgeRank (H.-D. Do & Do, 2023), AWP (T. Do et al., 2023)), and the tooling through which trust is formed in practice judges one contract or one account at a time (Human Passport, n.d.). Relations between contracts have been measured (Kitzler et al., 2023), but in the work reviewed here no ranking is computed from them on a graph whose nodes are contracts only; this gap sets the graph of every research question. Second, none of the reviewed methods uses the ERC-20 allowance, the on-chain record of a delegation of spending power, as a ranking edge. Approvals have served address clustering and security analysis (Sections 2.8 and 2.18), and it is unknown whether, as edges between contracts counted against what the owner holds, they order contracts differently from transfers and at what cost (O1 and RQ1). Third, the experiment in which a PageRank ordering reached important web pages sooner than other orderings (Cho et al., 1998) has, in the work reviewed here, no counterpart on a ledger: the ledger rankings are checked by rank correlation with statistics of their own edges, which shows coherence and not that a ranking leads to the contracts that users rely on (O2 and RQ2). Fourth, a walk is seldom compared out of window with the raw count that it smooths, although such counts are the standard baselines of link prediction (Liben-Nowell & Kleinberg, 2007) (O3 and RQ3). Finally, multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) have not, in the work reviewed here, been applied to two on-chain relations over one

set of contracts, and no study reports whether an allowance layer improves a transfer layer under rules fixed before the outcome is seen, as preregistration asks (Nosek et al., 2018) (O4 and RQ4, O5 and RQ5); the thesis fixed such rules for its registered analysis (Appendix 8.6) and applies them to the main analysis as a re-analysis. C-PR, the method of the thesis, is built for the first, second and fifth gaps; the third and fourth shape its evaluation.

2.23 Open problems motivating this research

The literature leaves these problems in on-chain reputation unresolved:

1. Selection: Among the very many contracts of a rollup, which should be looked at first when labels are sparse (D. Lin et al., 2024)?
2. Edge semantics: Which on-chain relations between contracts show that one relies on another (Kitzler et al., 2023)?
3. Evaluation: What shows that a ranking leads to the contracts that matter, when statistics of its own edges agree with it by construction (Cronbach & Meehl, 1955)?
4. Computational cost: As graphs grow, can scores still be refreshed at interactive latencies (Langville & Meyer, 2006)?

The thesis takes on (1)–(4) with a graph of contracts, a walk that couples its two edge types, the discovery-order test, out-of-window prediction against the raw counts, and runtime benchmarks. Problem (5) is analyzed at the model level in Chapter 5.

Chapter 3 turns this gap statement into an operational design.